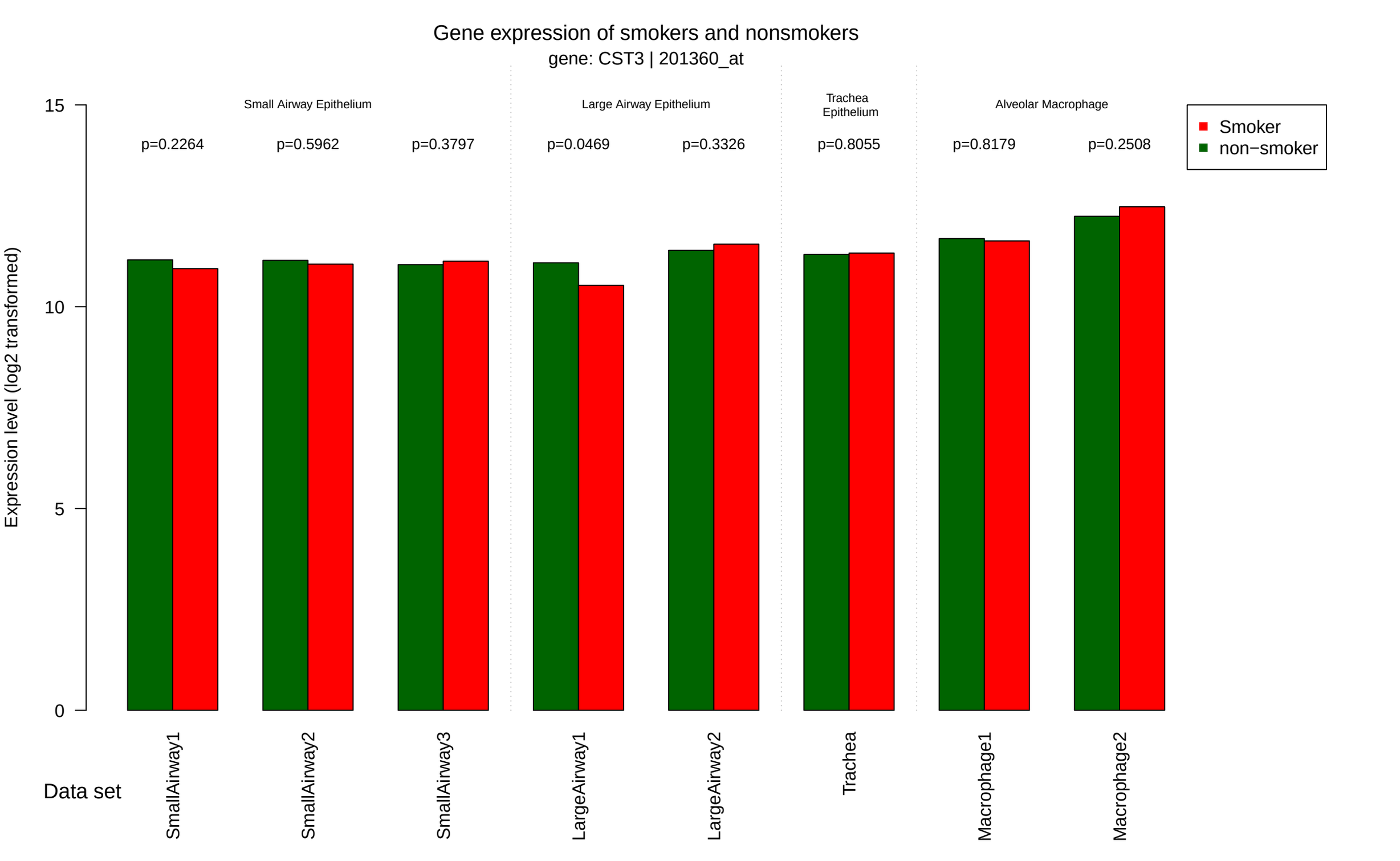
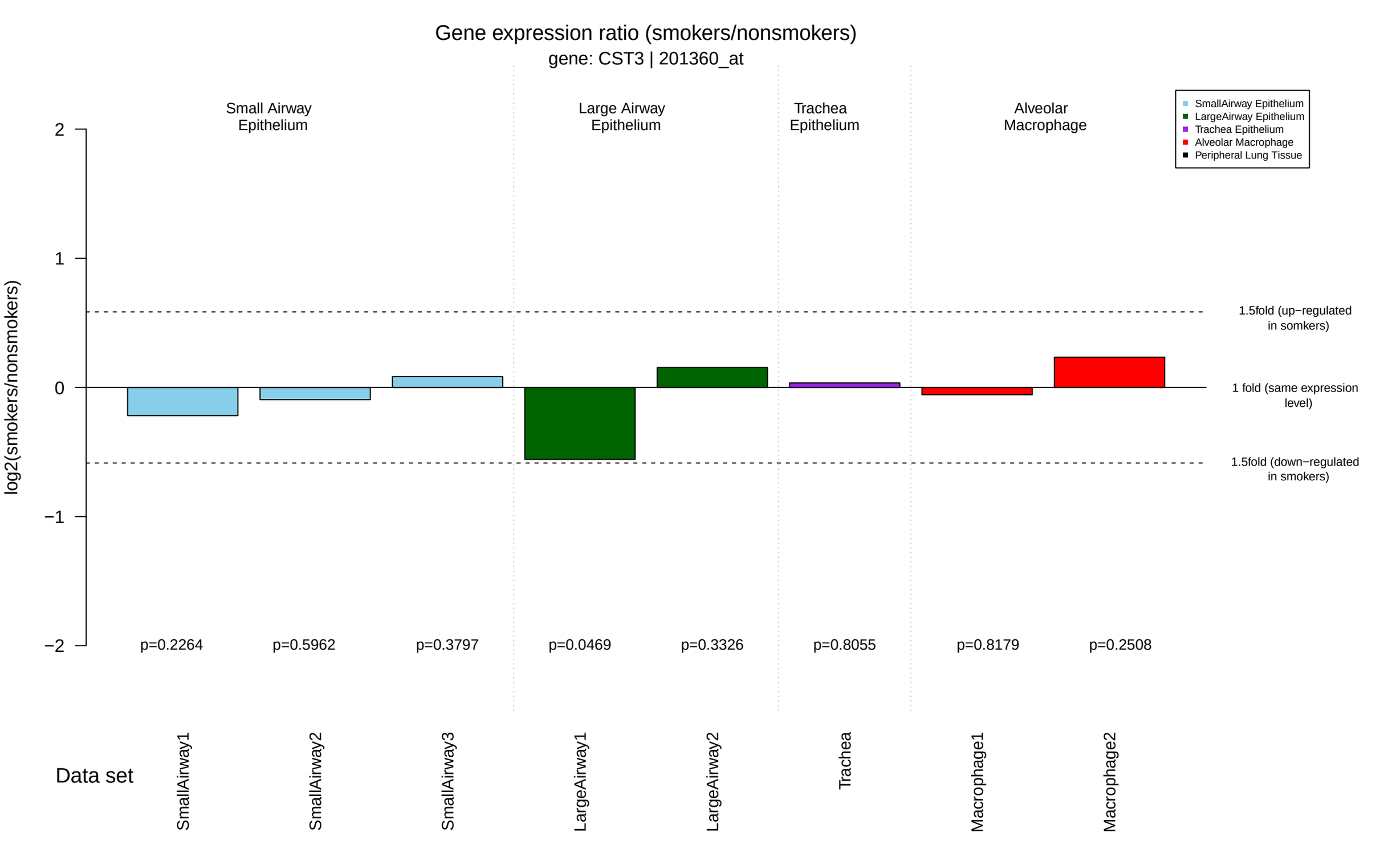
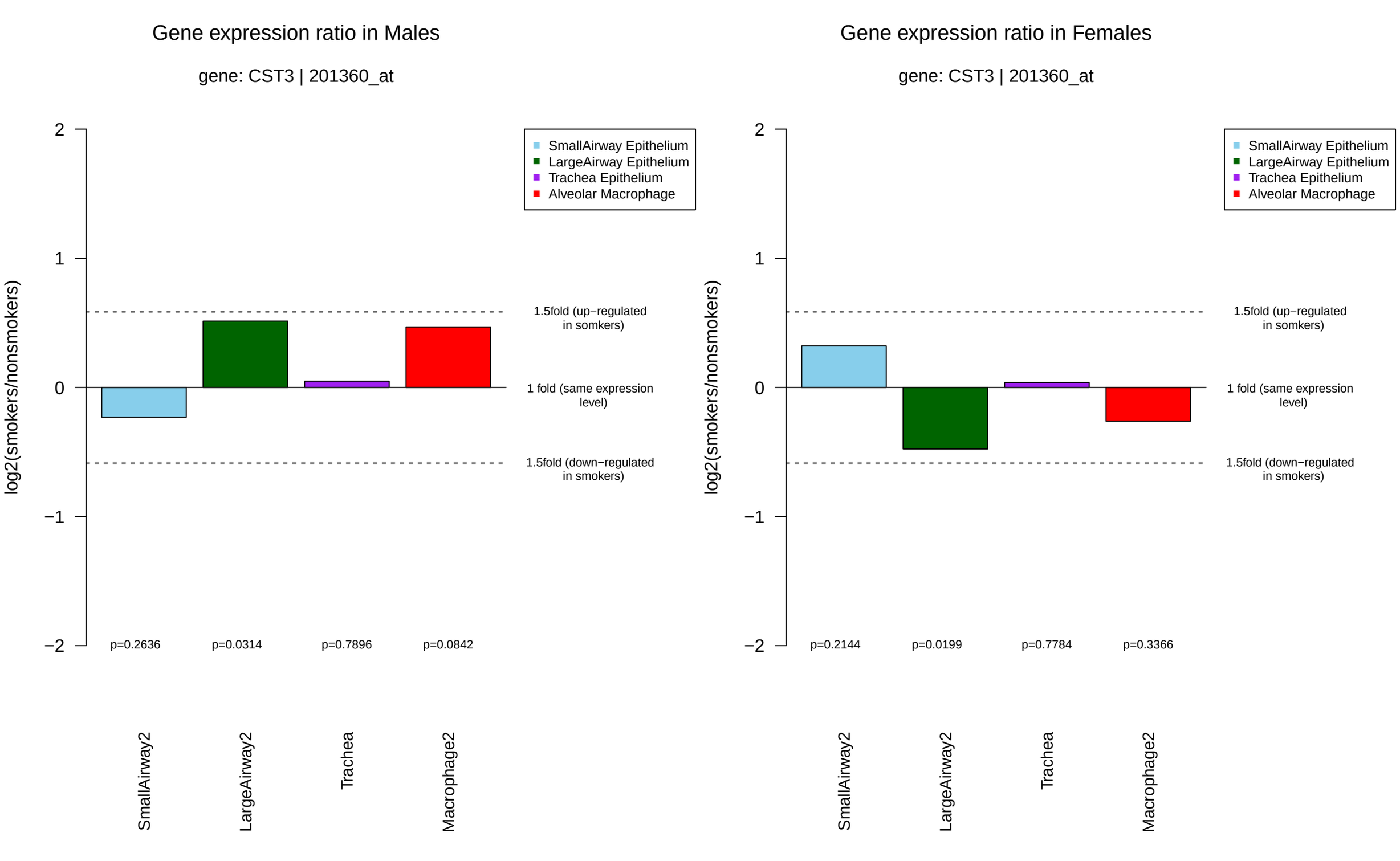
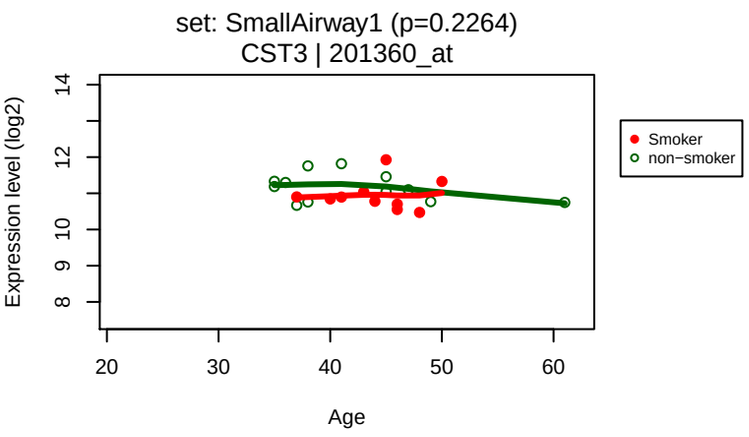
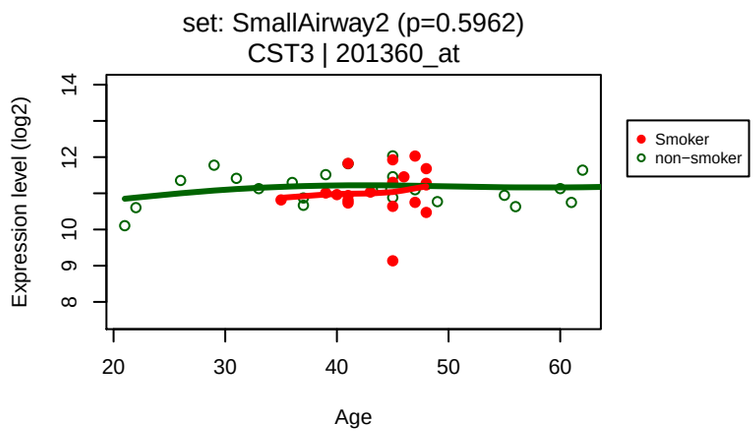
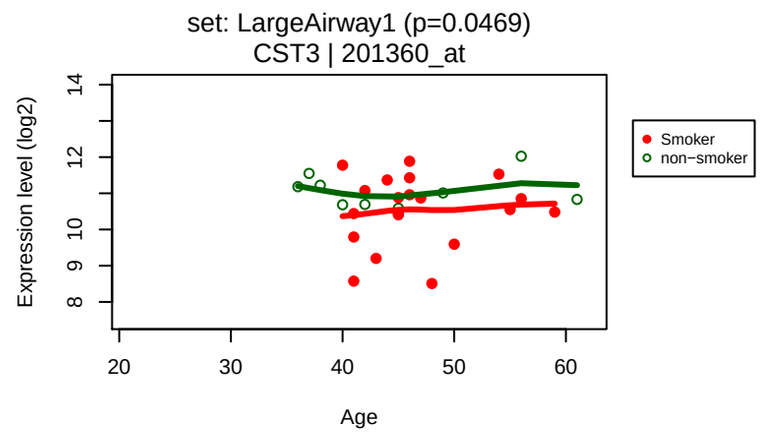

Gene expression of smokers and nonsmokers
gene: CST3 | 201360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2264
p=0.5962
p=0.3797
p=0.0469
p=0.3326
p=0.8055
p=0.8179
p=0.2508
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

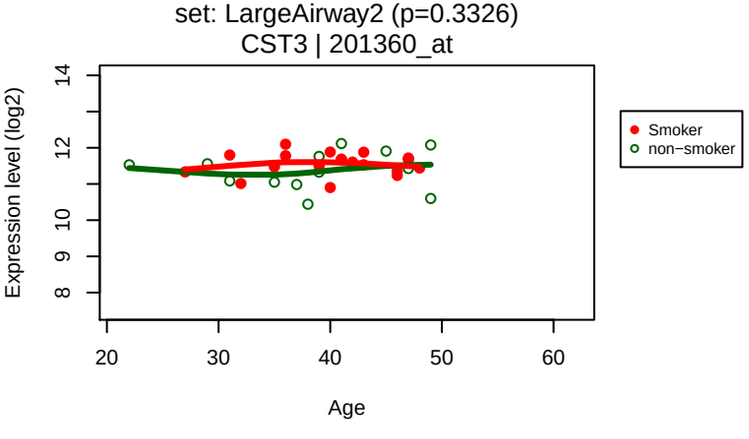

Gene expression ratio (smokers/nonsmokers)
gene: CST3 | 201360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2264
p=0.5962
p=0.3797
p=0.0469
p=0.3326
p=0.8055
p=0.8179
p=0.2508
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

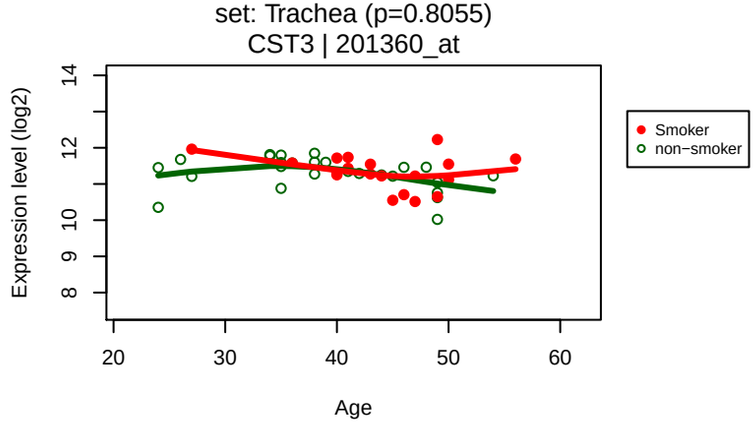

Gene expression ratio in Males
gene: CST3 | 201360_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2636
p=0.0314
p=0.7896
p=0.0842
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CST3 | 201360_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2144
p=0.0199
p=0.7784
p=0.3366
SmallAirway2
LargeAirway2
Trachea
Macrophage2

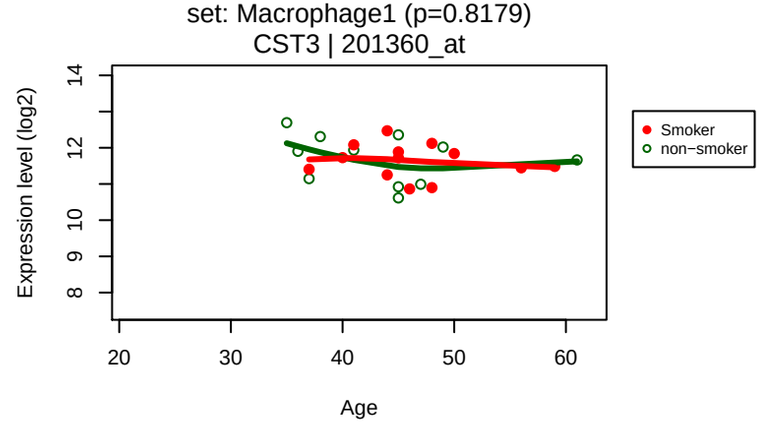

set: SmallAirway1 (p=0.2264)
CST3 | 201360_at
Expression level (log2)
Age
Smoker
non−smoker

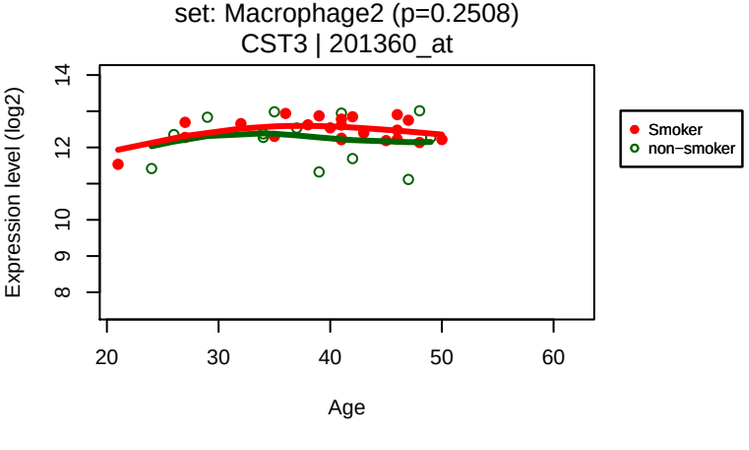

set: SmallAirway2 (p=0.5962)
CST3 | 201360_at
Expression level (log2)
Age
Smoker
non−smoker

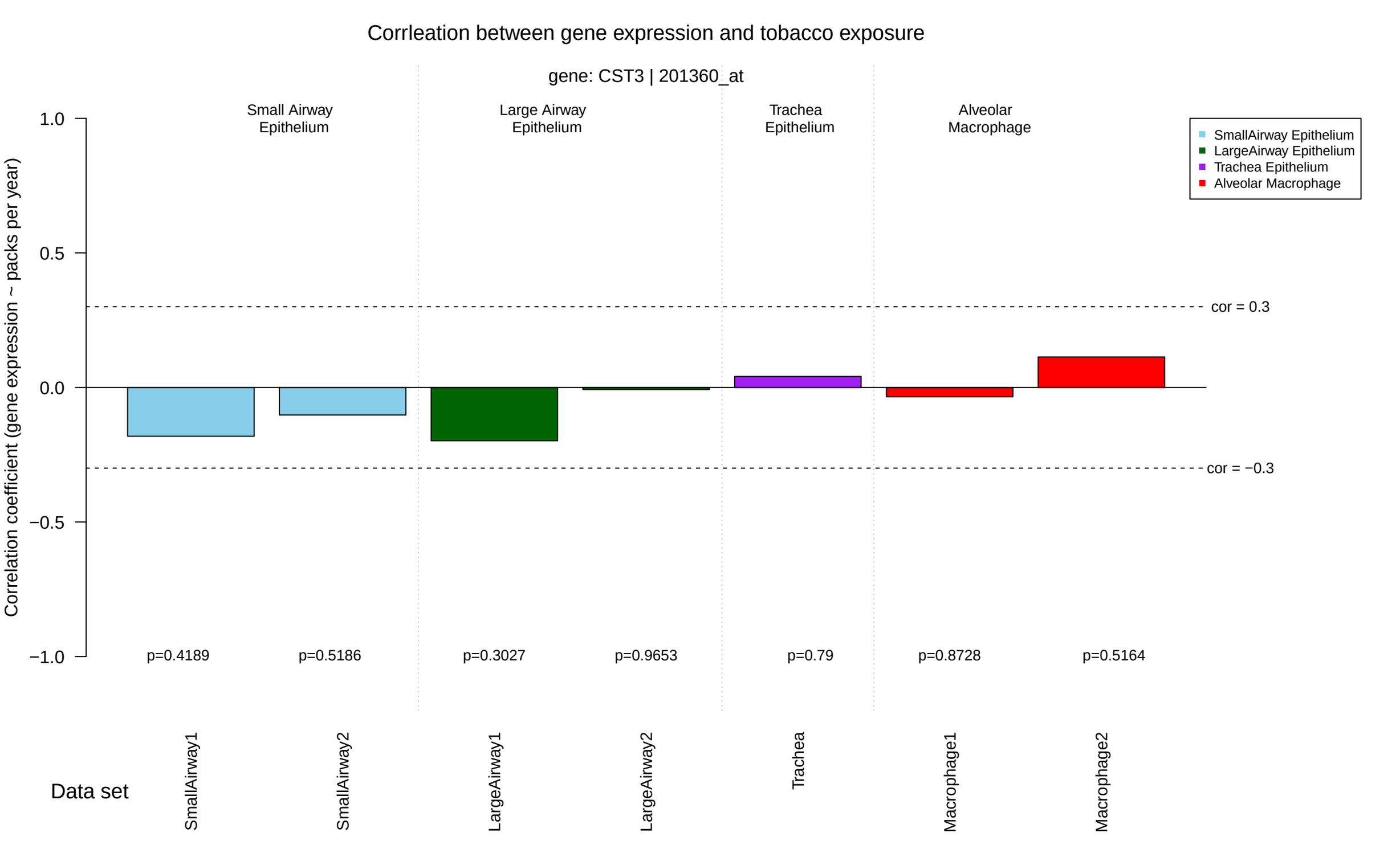

set: LargeAirway1 (p=0.0469)
CST3 | 201360_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3326)
CST3 | 201360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8055)
CST3 | 201360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8179)
CST3 | 201360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2508)
CST3 | 201360_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CST3 | 201360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4189
p=0.5186
p=0.3027
p=0.9653
p=0.79
p=0.8728
p=0.5164
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2